

Quiz4, MAT1375 Professor Chiu

ID: _____

Name: _____

- This quiz consists of 10 questions for a total of 10 points.
- You have 10 minutes to complete the quiz.
- Wishing you success.

(5 points) True or False questions:

1. T The parabola $y = x^2 + 3$ is the parabola $y = x^2$ shifted up by 3 units.
2. T The parabola $y = (x + 2)^2$ is the parabola $y = x^2$ shifted 2 units to the left.
3. T The parabola $y = -(x + 2)^2$ is the reflection of $y = (x + 2)^2$ about the x -axis.
4. T Let $f(x) = 3x + 6$ and $g(x) = 2x - 8$. Then the domain of $\left(\frac{f}{g}\right)(x)$ is $(-\infty, 4) \cup (4, \infty)$.
5. F Let $f(x)$ and $g(x)$ be two functions. Then $(f \circ g)(x) = (g \circ f)(x)$.

(3 points) 6. Let $f(x) = \frac{4}{x+2}$ and $g(x) = x^2 - 3x$. Find $(f \circ g)(x)$ and state its domain.

$$(f \circ g)(x) = f(g(x)) = \frac{4}{g(x)+2} = \frac{4}{x^2-3x+2} \quad \uparrow \quad \frac{4}{(x-1)(x-2)}$$

$$\begin{array}{r} x^2 - 3x + 2 = (x-1)(x-2) \\ x \quad -1 \\ x \quad -2 \end{array}$$

Since $x=1$ or $x=2$ will make the denominator of $(f \circ g)(x)$ zero, then $x=1, x=2$ are not included in the domain.

$$\Rightarrow D_{f \circ g} = \{x \mid x \in \mathbb{R}, \text{ but } x \neq 1, x \neq 2\}$$

Please turn over and finish the rest of the question.

(2 points) 7. Given two functions f and g by the tables:

Please use the information from the table of f and g to complete the following table:

x	1	2	3	4	5	6
$f(x)$	3	1	2	5	6	3

x	1	2	3	4	5	6
$g(x)$	5	2	6	1	2	4

x	1	2	3	4	5	6
$f(x)$	$f(1)=3$	$f(2)=1$	$f(3)=2$	$f(4)=5$	$f(5)=6$	$f(6)=3$
$(g \circ f)(x)$	$g(f(1))$	$g(f(2))$	$g(f(3))$	$g(f(4))$	$g(f(5))$	$g(f(6))$
$g(f(x))$	$= g(3)$	$= g(1)$	$= g(2)$	$= g(5)$	$= g(6)$	$= g(3)$
	$= 6$	$= 5$	$= 2$	$= 2$	$= 4$	$= 6$

End of this quiz.